

05

09 E II

05

Model Answers

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 மேலதிக வாசிப்பு நேரம் - 10 நிமிடங்கள்
Additional Reading Time - 10 minutes

Answer all questions on this paper itself.
(Each question carries 10 marks.)

- Do not
write
in this
column

- 2pt

- 4pt

- 1pt

- 2pt

- 1pt

- DNA / Circular DNA /
chloroplast DNA



- 1pt

- 1pt

d. Give two differences observed in the membrane system of chloroplast when compared with the membrane system of mitochondria

- In chloroplasts outer and inner membrane are smooth while in mitochondria outer membrane is smooth but inner membrane is convoluted
- Inside the chloroplasts there is another membrane system / chloroplasts consists of grana

2pt

(v) Some events that occur during meiosis of an animal cell are given below. State the phase of meiosis in which each event occurs

- Genetically non identical haploid daughter cells are formed Cytokinesis
- Homologous pairs move towards metaphase plate Prophase I
- Breakdown of proteins attaching sister chromatids Anaphase II

3pt

(vi) A laboratory experiment was carried out to determine the effect of temperature on the rate of enzymatic reaction by using amylase and starch solutions

a. State the reagent used to test the presence or absence of starch in this experiment

- I_2/KI solution

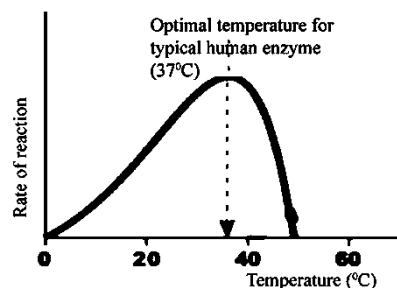
1pt

b. State the observation which indicates that starch has been completely broken down into maltose.

- Dark blue colour will not appear / the drop of mixture remains at brown colour/colour of I_2/KI solution

1pt

c. Draw the expected graph of 1/time against temperature obtained by carrying out the experiment at several different temperature



3pt

(vii) State the site of photosynthesis in eukaryotes and prokaryotes respectively

- Eukaryotes – Chloroplasts / stroma and grana of chloroplasts
- Prokaryotes – Take place on membrane (which show no stacking)

2pt

(viii) In which phase of Calvin cycle NADPH is required ?

- Reduction phase / Reduction of 1,3-Bisphosphoglycerate to Glyceraldehyde 3-phosphate

1pt

(B) (i) State the events in correct chronological order from the origin of the first cyanobacteria to the origin of chloroplasts

- Iron (Fe^{2+}) ion were oxidized.
- All of the dissolved iron has precipitated
- Additional O_2 dissolved in the water until water bodies became saturated with O_2 .
- The increase of photosynthetic bacteria contributed to the increment of the amount of atmospheric oxygen

4pt

- (ii) State the advantages of artificial classification despite the fact that natural classification is based on true relationships
- Easy to use, easy to expand by adding more groups
- (iii) Some invertebrate animals are listed below. Select the most suitable example/s from the given organisms for each of the following features

2pt

Earthworm, Sea cucumber, Pin worm, Jelly fish, Flukes

- a. Have no skeletal system Flukes
- b. Presence of Alimentary canal for the first time Pin worm
- c. Bisexual Earthworms, flukes
- d. No specialized excretory system Jelly fish, Sea cucumber, Pinworm (Any 2)
(Pinworms have longitudinal excretory ducts)

6pt

2. (A) (i) Explain how leaf orientation helps plants to prevent damage caused by excessive light intensity

2pt

- Some plants have vertically arranged leaves.
- light rays are parallel to the leaf surfaces (no leaves receive too much of light.)

- (ii) State how the energy required for accumulation of K^+ in guard cells is generated during the opening of stomata in day time

- Transfer of electrons during photosynthesis of the chloroplast in guard cells.
/ by light reactions in chloroplast of guard cells

1pt

- (iii) Define plasmolysis and explain how it occurs when a flaccid cell is placed in a solution with a higher solute potential than that of the cell

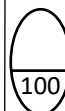
- The protoplast of the cell shrinks and pulls away from the cell wall. This is known as plasmolysis.
- Since the external solution has a lower water potential / more negative water potential, than the cell, water diffuse out of the cell.

2pt

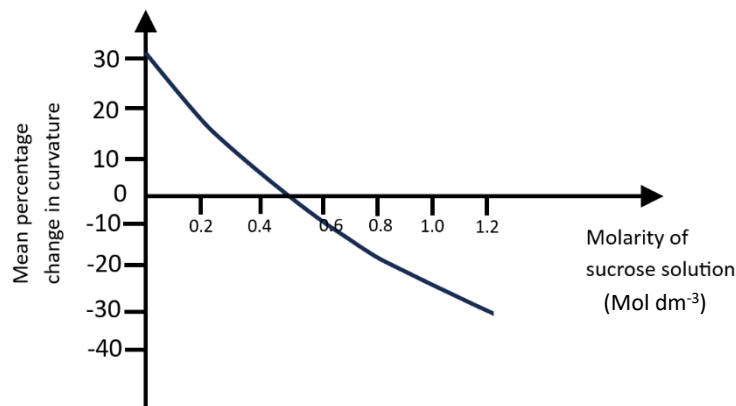
- (iv) Six *Alocasia* petioles, each 6 cm in length and of uniform diameter, were taken and each petiole was split radially into four equal strips. 20ml of sucrose solutions of different concentrations and other necessary apparatus were also provided. Write down the suitable experimental procedure to determine the water potential of the *Alocasia* petioles using the given materials.

- Place each piece on blank paper and measure the initial curvature angle.
- Immerse four such strips in each of the sucrose solutions and set aside with the lid closed for at least one hour
- (Remove the strips from the solutions.) Blot the excess solution using a blotting paper and place on a sheet of paper.
- Draw the outlines of each strip to record the curvature again & measure the angle.
- Plot a graph against the percentage change in curvature (in Y axis) verses concentrations of sucrose solutions (in X axis)
- Determine the concentration of the solution which caused no change in curvature from the graph
- Determine water potential using the given table

7pt



b. The graph plotted from the above experiment and the data table are given below



Molarity	Solute potential of solution (kPa)
0.2	-540
0.3	-820
0.4	-1120
0.5	-1450
0.6	-1800
1.2	-5560

1. Using the given graph, determine the molarity of the sucrose solution at which no change in curvature occurs0.5 mol dm⁻³..... 1pt

2. Calculate the water potential of *Alocasia* petiole using appropriate steps

- (Ψ_s of 0.5 mol dm⁻³ sucrose solution is -1450kPa)
- (Solution is open to atmosphere therefore), Ψ of sucrose solution is -1450kPa
- At equilibrium Ψ of sucrose solution = Ψ of *Alocasia* petiole
- Ψ of *Alocasia* petiole = -1450kPa

3pt

(B) (i) Give two reasons why blood is considered as a specialized connective tissue

- Matrix is not secreted by the blood cells
- Fibers are formed only during blood clotting

2pt

(ii) State the function of each of the following parts of a neuron

- Dendrites Receive nerve impulses from other neurons
- Axons Transmit impulses to other neurons, cells or muscles
- Myelin sheath Increase the speed of nerve impulse transmission

3pt

(iii) Give four functions of lipids in a balanced diet

- Provide energy and heat
- Helps in transport and storage of fat soluble vitamins / Vitamin A,D,E and K
- Store energy as fat
- Helps to synthesize steroid hormones from cholesterol
- Provide insulation (Any four)

4pt

(iv) Explain how lymph is transported through the lymph vessels

- By rhythmic contractions of lymph vessel walls
- And skeletal muscle contraction

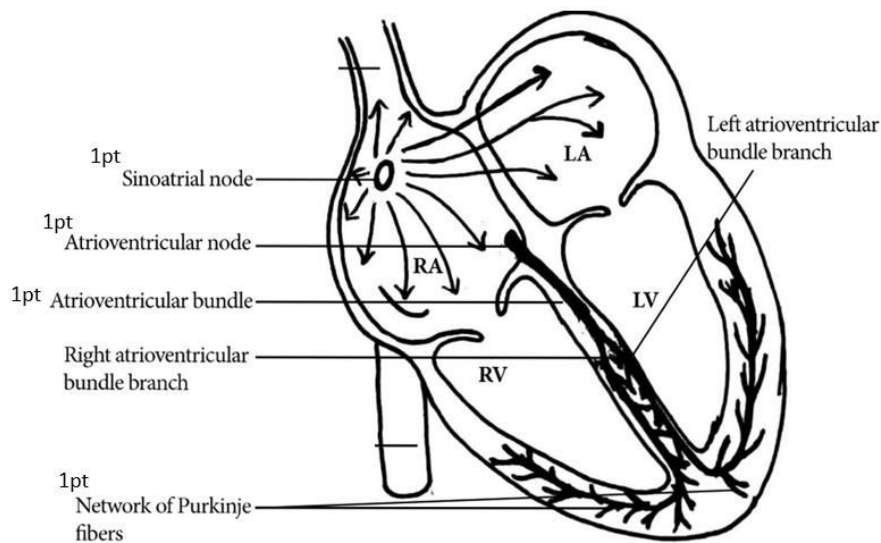
2pt

(v) State the location of semilunar valves in heart

- They are found at the point where the pulmonary artery
- and aorta leave the ventricles

2pt

- (vi) On the given outline of the heart, draw and clearly label the components of the cardiac conduction system including the nodes of specialized of neuromuscular cells and specialized conducting fibers



4pt

- (C) (i) State three ways in which ready-made antibodies can be administered for an individual to develop passive immunity

- Blood plasma or serum
- As injections of pooled human immunoglobulin
- As monoclonal antibodies.

3pt

- (ii) What are known as allergens

- Antigens that induce hypersensitive reactions in some persons are known as allergens

1pt

- (iii) State two typical symptoms of an allergic reaction in the human body after the entry of allergens

- Sneezing, runny nose, teary eyes, breathing difficulties (any 2)

2pt

- (iv) State one disease caused by the antibodies mistakenly produced by the immune system against the body's own tissues.

- Rheumatoid arthritis

1pt

3. (A) (i) Name the structure that connects the two cerebral hemispheres together

- Corpus callosum

1pt

- (ii) Name the part of the brain that forms bridge between two hemispheres of cerebellum

- Pons Varolii (Nerve fibers of Pons Varolii)

1pt

- (iii) State three functions of midbrain

- Acts as relay stations for ascending and descending nerve fibers
- Receives and integrates sensory information (auditory and visual)
- Coordinates auditory and visual reflexes

3pt

Do not
write
in this
column

100

(iv) State the special type of receptors that detect the following stimuli

- a. Light touchMerkel discs 3pt
- b. Extreme pressure and temperature Pain receptors
- c. Low peripheral temperature Krause end bulbs 3pt

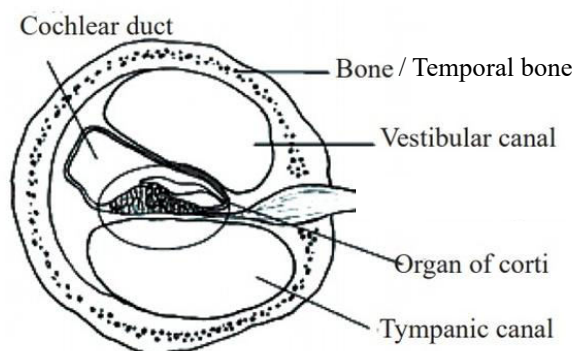
(v) State the location of yellow spot in the human eye

- At the center of posterior part of retina 1pt

(vi) What is known as binocular vision in human

- Seeing the visual field using two eyes
- with greater overlapping of field of view is called binocular vision 2pt

(vii) Draw the structure of cross section of cochlea of human ear



5pt

(B) (i) State the functions of following hormones

- a. ACTH Stimulates secretion of adrenal cortex hormones / Glucocorticoid
- b. PIH Inhibits the secretion of prolactin hormone (from anterior pituitary)
- c. ADH Stimulates reabsorption of water by increasing permeability to water

3pt

(ii) Describe the location and pathway of the vas deferens from its origin to the urethra in the human male reproductive system

- Each vas deferens starting from each epididymis
- extends around and behind the urinary bladder,
- where it joins a duct from a seminal vesicle,
- forming a short ejaculatory duct. (which opens into urethra) 4pt

(iii) Consider the following cells related to spermatogenesis and oogenesis in human and state the ploidy level and the number of chromosomes present in each cell

- a. Spermatogonial stem cell Diploid / $2n$, 46 / 23pairs
- b. Secondary spermatocyte Haploid / n , 23
- c. Primary oocyte Diploid / $2n$, 46 / 23pairs

6pt

(iv) State three functions of placenta

- The placenta supplies oxygen and nutrients to the fetus and excrete waste products
- Provide immune protection to the developing embryo/fetus.
- The placenta produces hormones (e.g. hCG, progesterone etc.) needed to sustain the pregnancy. 3pt

(C) (i) What is meant by polyallelism

- Polyallelism refers to the presence of multiple alleles for a single genetic locus,
- a phenomenon where certain traits are determined by the combination of more than two types of alleles.

2pt

(ii) A father with blood group B and a mother with blood group A can produce children with blood groups A and AB only, in equal proportions

a. Determine the genotypes of father and mother

- Father $I^B i$, Mother $I^A I^A$

2pt

b. When a woman who is the daughter of above parents marries a man with blood group O, none of their children will have blood group O. Also, the woman does not have Rhesus factor in her blood

1. Determine the blood group of the above woman

- AB^- / AB negative blood group

1pt

2. Determine the probability for having a son with blood group A for the above couple

(Probability of blood group A = $\frac{1}{2}$, Probability of male child = $\frac{1}{2}$)

- Probability for son with blood group A = $\frac{1}{4}$

1pt

3. The first son of the above couple is Rhesus positive (Rh^+), Explain the consequences if the mother becomes pregnant for the second time with a Rhesus positive fetus

- The anti Rh antibodies developed in her plasma in response to the first child's red blood cells
- can pass across the placenta to the fetus and destroy fetal red cells

2pt

4. (A) (i) What are known as recombinant DNA molecules

- DNA molecules made by laboratory methods of genetic recombination
- Which brings together DNA from different sources
- creating sequences that are not found naturally.

3pt

(ii) Cloning vector, gene of interest and required enzymes are provided, outline the steps involved in the preparation of recombinant vector

- The gene of interest is cut with a restriction enzyme
- The vector / plasmid / viral DNA should also be cut with the same restriction enzyme.
- These two should be mixed and allowed to integrate
- and ligated using DNA ligase

4pt

(iii) State the self-replicating unit that can function as a cloning vector in a eukaryotic system.

- Yeast artificial chromosomes / YAC / plasmid of yeast

1pt

100

- (iii) Two marker genes are commonly used to determine the success of transformation of host cells with recombinant vectors. State the primary function of each marker gene.

First marker / Selectable marker

- Colonies originating from the transformed cells can be screened among many untransformed cells. / Both transformed and untransformed cells will make colonies in suitable medium, they can be distinguished using selectable markers

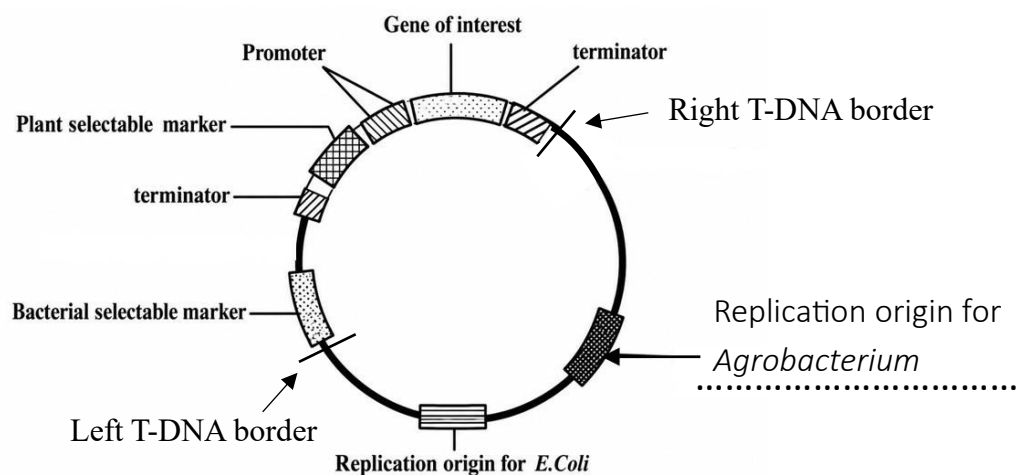
1pt

Second marker

- Distinguish the colonies with the vector containing the insert from those with vector only (without an insert)

1pt

- (iv) A labelled diagram of Ti plasmid vector which is used in DNA delivery systems is given below



1pt

- a. What is meant by transfer DNA

- A part of this plasmid (with the gene of interest) can actually be transferred to plant genome, and hence called transfer DNA or T-DNA.

1pt

- b. Label the left and right border sequences of the T-DNA in the above diagram

1pt

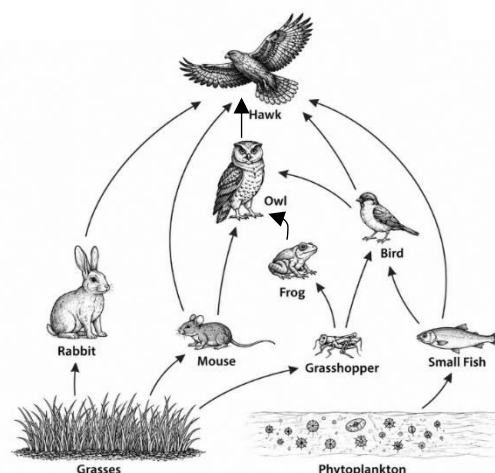
- c. Identify and write down the name of the unlabeled part in the diagram

- (B) (i) What is a food web

- The food web is an interconnected feeding relationships in an ecosystem.

1pt

- (ii) A food web which belongs to a lake-side ecosystem is given below



a. Identify one organism from the food web that represents each of the following roles

1. Organism with the greatest number of food sources Hawk 3pt
2. A primary carnivore Owl / Hawk / frog / bird
3. A tertiary consumer Owl / Hawk

b. Identify the least energy-efficient food chain from the given food web

- Grasses → Grass hopper → Frog → Owl → Hawk or 1pt
- Grasses → Grass hopper → Bird → Owl → Hawk

c. If the amount of energy fixed at the primary producer of the above least efficient food chain is $4 \times 10^{-2} \text{ J m}^{-2} \text{ day}^{-1}$, calculate the approximate amount of energy available at the highest tropic level

- $4 \times 10^{-2} \text{ J m}^{-2} \text{ day}^{-1} \times 10/100 \times 10/100 \times 10/100 \times 10/100 = 4 \times 10^{-6} \text{ J m}^{-2} \text{ day}^{-1}$ 2pt
- $4 \times 10^{-6} \text{ J m}^{-2} \text{ day}^{-1}$

d. State one function of phytoplankton other than being a primary producer

- (They live in warm seas and) very useful to maintain oxygen and carbon dioxide balance 1pt

e. State two major adverse effects caused by the destruction of phytoplankton due to increased UV radiation resulting from ozone layer depletion

- Reduce the CO_2 absorption capacity of oceans / increase the global temperature 2pt
- reduce the composition of food web in the sea ecosystem

(C) (i) What is meant by invasiveness

- It is the ability of pathogens to invade tissues by overcoming host's defense mechanisms 2pt
- And multiply for colonizing

(ii) The following enzymes are produced by pathogenic microorganisms. State the role of each enzyme in the pathogenicity of the microorganism

- a. Phospholipase Destroy animal cell membranes 3pt
- b. Hyaluronidase Destroys the body tissue by breaking down the hyaluronic
- c. DNase Alter the host's metabolic processes (by breaking down of host's DNA)

(ii) Identify whether each of the following processes mediated by microorganisms occurs under aerobic conditions, anaerobic conditions, or both

- a. Retting Both aerobic and anaerobic 4pt
- b. Composting of solid waste Aerobic
- c. Biogas production Anaerobic
- d. Vinegar production Both aerobic and anaerobic

(iii) What is the main reason for occurrence of Eutrophication in freshwater bodies

- Accumulation of excessive amounts of nitrates and phosphates 2pt
- that come from agricultural fertilizer / household chemicals such as detergents

(iv) Describe two external features that can be used to distinguish *Aedes* mosquito from *Culex* mosquito

- *Aedes* is Dark in colour *Culex* is Grey-black in colour
- *Aedes* have white markings/bands on their body, *Culex* do not have them
- In *Culex* nervures of wings have brown or blackish scales, *Aedes* do not have them

3pt

(v) State how the larval stage of the parasite that causes lymphatic filariasis enters the human body

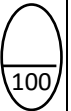
- (When the mosquito suck blood from human)
- Larval stages of parasite fall on to the skin (no marks for- into the skin)
- Then they penetrate the skin through the bite wound

2pt

(vi) What is the name of the condition caused by the accumulation of fluid in the scrotum due to obstruction of lymphatic vessels by adult filarial worms

- Hydrocele

1pt



1.	3	6.	1	11.	5 (C)
2.	2	7.	2	12.	5 (C)
3.	2	8.	3	13.	2
4.	3	9.	2	14.	3
5.	2	10.	5	15.	4

1. Cell wall is an extracellular structure of plant cells.
2. The chemical composition of the cell wall greatly varies from species to species and even from one cell type to another even in the same species.
3. In Plants, cell wall is generally made up of cellulose,
4. pectin
5. and hemicellulose.
6. Lignin
7. and suberin are found in some plant cells.
8. Plants generate two types of cell walls: primary and
9. secondary walls.
10. Young cells first secrete primary cell wall:
11. it is the wall laid down during plant cell division.
12. Due to the deposition of hardening substances on the primary wall a secondary cell wall is generated secondarily.
13. Primary cell wall is permeable,
14. relatively thin,
15. flexible,
16. composed mainly of cellulose fibers
17. and deposited on the middle lamella.
18. Water can move freely through the free spaces of cell wall.
19. Secondary cell wall lies between plasma membrane and primary cell wall.
20. It contains several layers of hard materials, forming a rigid structure.
21. In addition to cellulose, impermeable substances such as lignin and suberin are also incorporated in to the secondary wall.
22. Lignin cement anchors cellulose fibers together providing hard and rigid matrix,
23. giving the cell wall an extra support.

24. Just outside the primary wall there is a thin layer called middle lamella
25. which is rich in sticky polysaccharides called pectins / magnesium and calcium pectate
26. Middle lamella glues adjacent cells together.
27. Cell wall has pits through which cytoplasm of adjoining cells join
28. through plasmodesmata.
29. Plasmodesmata are microscopic channels which runs through plant cell walls.
30. They are cytoplasmic living connections between cytoplasm of adjoining cells.
31. These are membrane lined channels

Functions

32. Protection and support
33. Allows development of turgidity when water enters the cell
34. Prevents bursting during turgidity
35. Limits and control cell growth
36. Component of apoplast pathway
37. Maintaining cell shape
38. Hold the plant up against the force of gravity

Total 38 points

Any 37 points x 4 marks = 148 marks

If more than 37 points written, add two marks

Total 150 marks

6. (a) Briefly explain the process of fertilization of egg and sperm nuclei in Anthophytes

1. The pollen grain germinates after it is placed on the stigma.
2. It extends a pollen tube that grows down through the style of the carpel.
3. The nucleus of the generative cell divides forming two sperms.
4. When the pollen tube reaches the ovary, it moves through the micropyle of ovule
5. and discharges two sperm into the embryo sac / female gametophyte
6. One sperm fuses with the egg cell
7. forming a diploid zygote
8. and the other sperm nucleus fuses with the two polar nuclei.
9. This type of fertilization is called double fertilization
10. After double fertilization, the ovule matures into a seed.
11. The zygote develops into the embryo.
12. The triploid nucleus develops into the endosperm that store food.
13. The significance of double fertilization is that it synchronizes endosperm development with the development of the embryo.
14. If fertilization does not occur that prevents plants from wasting nutrients on infertile ovules.

6. (b) Describe the values biodiversity

1. Environmental service value:
2. Carbon dioxide fixation through photosynthesis,
3. maintaining of essential nutrition cycles,
4. maintaining water cycle and recharging of ground water,
5. soil formation and protection from erosion,
6. regulating climate by recycling moisture into the atmosphere,
7. water purification,
8. pollination.

9. Recreational value:
10. Natural landscapes at undisturbed places are a delight to watch
11. and also provide opportunities for recreational activities and hobbies such as bird watching, photography etc.
12. Biodiversity provides inspiration in artistic activities like poetry, painting, dance etc.
13. It promotes eco- tourism, helps to generate revenue by designing of zoological, botanical gardens, national parks etc.
14. Ethical value:
15. This is the right of all living beings to live on this planet,
16. humans have right to decide which species should exist since we are just a small part of the greater creation of nature.
17. Educational/Scientific value:
18. knowledge about biodiversity helps in new scientific discoveries and technological innovations to find solutions to the problems we face today.
19. E.g. learning of other animals like nematodes, rats and primates has helped in understanding human body
20. and development of medicines,
21. knowledge about how animals react before a natural disaster is helpful in disaster management,
22. Interacting with biodiversity is proven to be helpful in developing creativity, relieving stress and development of personality.
23. Social/Cultural/Religious values:
24. E.g. Some wetland sites are sacred to Aborigines of Australia,
25. Twenty eight species of trees are sacred for Buddhists,
26. Bulls are considered to be an important part of Hindu culture.

14 points + 26 points = 40 points

Any 37 points x 4 marks = 148 marks

If more than 37 points written, add two marks

Total 150 marks

7. (a) Describe the characteristic features of phylum Chordata

1. Longitudinal,
2. flexible rod called
3. notochord
4. located between digestive tube and nerve cord.
5. It is extending from anterior to posterior
6. providing support in at least embryonic stage.
7. Dorsal, hollow,
8. single nerve cord
9. located dorsal to the notochord.
10. In all chordate embryos there are slits or clefts in pairs either side of pharynx
11. that opens to the outside of body.
12. In terrestrial, adult chordates it disappears and remains in the aquatic adults and larval forms of terrestrial chordates
13. as respiratory structures.
14. Muscular tail that extends posterior to the anus present in the embryonic stages.
15. In some terrestrial adults it is reduced.
16. Ventral muscular heart is present in most chordates.

7. (b) Write an account on DNA polymerases and explain the DNA repair mechanism

1. There are several types of DNA polymerases.
2. One type of DNA polymerase initiates and DNA polymerization by adding a deoxyribonucleotide to the 3' end of a primer
3. continues polymerization on 5' to 3' direction
4. elongating the new DNA strand with deoxyribonucleotides with bases complementary to template
5. This DNA polymerase is almost 100% error-free in adding correct complementary nucleotides to growing strand
6. However one error in 10^5 added nucleotides is possible, has a proofreading mechanism and therefore can correct its own mistakes
7. and the error rate will be reduced by 100,000 times to one in 10^{10} .
8. When a wrong nucleotide is added to the growing chain of DNA, by DNA Polymerase this mismatch is identified by the same enzyme,
9. stops adding the next nucleotide,
10. removes the incorrect nucleotide by exonuclease activity
11. and continues again its polymerase activity.
12. Another type of DNA polymerase identifies DNA-RNA hybrid, removes ribonucleotides and replaces those with deoxyribonucleotides
13. and so replaces the RNA primer with DNA.

DNA repair

14. Distorted locations are usually recognized and repaired by the DNA repair mechanism of the cell before it becomes permanent,
15. and that reduces the accumulation of mutations.
16. Large number of DNA repairing enzymes are found in various organisms.
17. Such enzymes may cut off the mismatched sequences in the damaged strand
18. and replace with correct nucleotides.
19. The cutting (excision) is done by a nuclease
20. and filling the gap, using the correct strand as the template, is done by a type of DNA polymerase.
21. This is called nucleotide excision repair.
22. DNA ligase completes the DNA strand by sealing it
23. with a phosphodiester bond.

16 points + 23 points = 39 points

Any 37 points x 4 marks = 148 marks

If more than 37 points written, add two marks

Total 150 marks

8. Explain the role of hypothalamus and kidneys in maintaining osmotic balance in the human body

1. When there is osmotic balance, amount of water and concentration of solutes is same inside and outside the cells.
2. Osmoregulation is important for organisms to keep a constant, optimal osmotic pressure within the body.
3. In the humans, osmoregulation ensures that the total blood volume
4. and the concentration of dissolved substances in the plasma and tissue fluids remain constant within a favourable range.
5. In the human body, osmotic balance is achieved by two ways: controlling the amount of water
6. and controlling the amount of solutes gained and lost by the body.
7. Blood water homeostasis is controlled by the hypothalamus
8. Hypothalamus contains osmoreceptors which can detect the osmolarity of the blood passing through the brain.
9. Hypothalamus - Controls the sensation of thirst
10. And secretion of the hormone ADH from the posterior pituitary.
11. When blood osmolarity is increased beyond the physiological limits, it is sensed by the osmoreceptors in the hypothalamus
12. Hypothalamus generates thirst,
13. Drinking water reduces blood osmolarity
14. Blood osmotic pressure returns to normal (300 mOsm/L)
15. Thereafter the Osmoreceptors in hypothalamus is no longer stimulated the stimulation stops by negative feedback mechanism
16. Hypothalamus stimulates the posterior pituitary to release ADH to the blood circulation
17. Increases secretion of ADH acts on the kidney tubules
18. stimulates the reabsorption of water through distal convoluted tubules of the nephrons
19. and collecting ducts
20. The increased reabsorption of water producing concentrated urine.
21. So the Blood osmotic pressure returns to normal (300 mOsm/L)
22. Thereafter the Osmoreceptors in hypothalamus is no longer stimulated the stimulation stops by negative feedback mechanism
23. and ADH is not released further
24. When blood osmolarity is decreased, ADH is not secreted,
25. water reabsorption through distal convoluted tubules of the nephrons and collecting ducts stop
26. thereby producing diluted urine.
27. Similarly, kidneys also play a major role in osmoregulation in the human body.
28. Kidneys regulate Blood pressure and volume by renin angiotensin- aldosterone system
29. When the Blood pressure or blood volume drops (for example, due to dehydration or blood loss)
30. The sensors in the kidney detects the decrease in blood pressure or volume
31. It stimulates kidney to produce and secrete the enzyme Renin
32. Angiotensinogen From liver is converted to Angiotensin I by the renin
33. Then the Angiotensin I is converted to Angiotensin II (by the catalytic action of Angiotensin Converting Enzyme)
34. In addition, low blood volume and low blood sodium ions also stimulate the kidneys to produce Angiotensin II

35. Angiotensin II also stimulates the adrenal cortex of Adrenal gland
36. to secrete aldosterone hormone
37. Aldosterone stimulates the reabsorption of more sodium ions and H₂O by the kidney tubules
38. Which is accompanied by the water retention,
39. thereby increasing blood volume and blood pressure to Normal

Total = 39 points

Any 37 points x 4 marks = 148 marks

If more than 37 points written, add two marks

Total 150 marks

9. Briefly describe the role of microorganisms in cycling of elements in biogeochemical cycles

Carbon cycle

1. Photosynthesis is the important first step in carbon cycle, in which, the atmospheric inorganic carbon dioxide is reduced/fixed to form organic compounds
2. Photoautotrophs such as plants, cyanobacteria, algae and photosynthetic bacteria fix carbon dioxide using energy from sunlight.
3. Chemoheterotrophs such as animals and protozoa, depend on organic compounds produced by autotrophs to utilize them as their carbon source.
4. Through food chain, carbon fixed from carbon dioxide by autotrophs, transferred from organism at lower trophic levels to the organisms at higher trophic levels.
5. Both autotrophs and chemoheterotrophs, release a part of their fixed carbon as carbon dioxide to the atmosphere
6. through respiration.
7. This carbon dioxide is again made available for autotrophs.
8. In chemoheterotrophs, undigested food is released to the environment as faeces
9. which is later decomposed by soil microorganisms.
10. Rest of the carbon fixed in organisms, remain within them until they die.
11. Once the organisms are dead, these organic compounds are decomposed and carbon dioxide is returned back to the atmosphere.
12. Microorganisms, mainly bacteria and fungi play a major role in organic matter decomposition.
13. Carbon cycle in relation to methane gas - Ocean sediments contain a large amount of methane.
14. However, about 80% of methane generated within ocean is consumed by microorganisms called methanotrophs (before it reaches to the atmosphere)
15. Despite of the above, methanogenic bacteria in the ocean's depths are constantly producing more methane.

Nitrogen cycle

16. About 80% of molecular nitrogen available in the atmosphere.
17. Therefore, it is essential to convert that atmospheric molecular nitrogen into bioavailable forms of nitrogen.
18. There are four key steps in the nitrogen cycle - Ammonification,
19. Nitrification,
20. Denitrification
21. and nitrogen fixation.

22. Ammonification - More than 90% of organic nitrogen in the soil exist as proteins.
23. Proteins from dead plants and animals are decomposed
24. by extracellular proteolytic enzymes secreted by microorganisms into amino acids.
25. Resulting amino acids are taken into microbial cells and are then subjected to ammonification
26. Amino groups of amino acids are converted into ammonia (NH₃).
27. In moist soil, ammonia is solubilized in water to form ammonium ions (NH₄⁺).
28. Nitrification is the process of oxidation of nitrogen in the ammonium ion to produce nitrate.
29. In first stage, microorganisms such as *Nitrosomonas*,
30. oxidizes ammonium ions (NH₄⁺) into Nitrites (NO₃⁻)
31. In the second stage, microorganisms such as *Nitrobacter*,
32. oxidizes Nitrites (NO₃⁻) into Nitrates (NO₂⁻)
33. In the absence of atmospheric oxygen, some microorganism use nitrate and reduces it into N₂
34. Denitrification causes loss of nitrogen into the atmosphere. It reduces available nitrogen in the soil.
35. Denitrification frequently occurs in waterlogged soils, where limited oxygen is present.
36. *Pseudomonas* converts nitrate ions into molecular nitrogen through following steps
37. nitrate to nitrite,
38. nitrite to nitrous oxide
39. and nitrous oxide to nitrogen gas
40. The process of conversion of nitrogen gas to ammonia is termed as nitrogen fixation.
41. Bacteria that are able to fix nitrogen require nitrogenase enzyme.
42. Nitrogenase enzyme is inactivated by oxygen.
43. There are two types of nitrogen fixing microorganisms, free-living and symbiotic.
44. Eg – Azotobacter, Nostoc, Clostridium (Any one)
45. Azola fixes nitrogen symbiotically with Anabaena and Lichen, a combination of fungus and an algae or cyanobacteria also fixes nitrogen. (Any one)

Total = 45 points

Any 37 points x 4 marks = 148 marks

If more than 37 points written, add two marks

Total 150 marks

10. Write short notes on the following.

(a) Chronic kidney disease of uncertain etiology (CKDu)

1. It is a condition of gradual loss of kidney function over time.
2. The root cause of CKDu has not been definitively established yet
3. Conventional risk factors are diabetes and high blood pressure,
4. genetic disorders
5. and urinary tract problems
6. The onset of the disease appears to be asymptomatic, and by the time the patient seeks treatment the kidneys have reached a stage of irreversible damage -End stage renal disease (ESRD)
7. Hypothesized reasons for CKDu. - Exposure to Heavy metal/ metaloid such as Arsenic (As) and Cadmium (Cd) through food and water.
8. Usage of low quality utensils for preparation of foods.
9. High Flouride (F) levels in water.
10. Exposure to pesticides.
11. Genetic factors.
12. Malnutrition and dehydration.

(b) Main types of joints in human skeletal system

1. Ball and socket joints
2. In these joints, ball shaped head is connected
3. with the cup shaped socket
4. and allows for wide range movements
5. such as flexion, extension, adduction, abduction, rotation and circumduction.
6. There are two ball and socket joints available in the human body: Shoulder joint
7. and hip joint.
8. Hinge Joints
9. The articulating ends of the bone fit together in such a way so it looks like a hinge of a door.
10. This allows only restricted movements such as flexion and extensions.
11. Examples for hinge joints are elbow joint, knee joint, ankle joint and joints between the phalanges of the fingers and toes.
12. Pivot joints
13. One bone fits into a hoop shaped ligament that holds it close to another bone
14. and allows it to rotate in the ring formed by the ligament.
15. These joints allow a bone or limb to rotate.
16. For example head rotates by the pivot joint formed by the atlas vertebrae within the transverse ligament ring and odontoid process of the axis

(c) Nanomedicine

1. Titanium dioxide (TiO₂) and silver (Ag) nanoparticles are used for sterilization of operation theatres and surgical instruments in hospitals.
2. These nanoparticles are able to destroy the microbes.
3. Silver nano lotions are applied in theatres to prevent the entry of microbes.
4. Nano particles are used in the production of antimicrobial coatings and nano - filters.
5. These nano filters are used to examine SARS patients.
6. Nano device sensors are used to monitor blood pressure, blood oxygen levels and hormone concentrations.
7. Scientists have developed “smart drugs” using tiny nanoparticles such as gold particles
8. that would destroy the cancer cells in a silent manner with very few side effects
9. Nano shells are used in the cancer treatment
10. Gold nano shells are used in bio imaging enhancements as well.
11. A Nano device is attached to the body which can release the required dosage of insulin at required time intervals in the treatment of diabetes
12. The fabrication of nano-formulated liposomes to deliver drugs for pain therapy
13. Since nanocarrier systems can be easily transferred to the airways, many respiratory diseases can be treated using nanotechnology
14. Viva gel is a vaginally applied microbicide in development for prevention of HIV and HSV
15. Nanocomposites are used to replace broken bones and to fill teeth as well

12 points + 16 points + 15 points = 43 points

Any 37 points x 4 marks = 148 marks

If more than 37 points written, add two marks

Total 150 marks